Theory of Computation

Practice Problems

Lecture 37

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1.) $r = 0^* + 1^*$

$s = 0^*1 + 1^*0$

$L(r) \cup L(s)$

$|w| \leq 5$

<u>Ans:</u> 19

Strings = ?

$L(r) = \{\varepsilon, 0, 0^1, 0^2, 0^3, 0^4, 0^5, 1, 1^2, 1^3, 1^4, 1^5\}$ 11

$L(s) = \{\cancel{\varepsilon}, \cancel{\phi}, 01, 0^21, 0^31, 0^41, 10, 1^20, 1^30, 1^40\}$ 8

2.) $L = b^* a (b^* a b^* a)^*$

$|w| \leq 3$

#f strings in $\overline{L}$

2 or 4

$L = \{a, aaa, ba, bba\}$

(2) $L_1 = b^* a b^* (a b^* a b^*)^*$

$L_2 = \{ w \in (a+b)^* \mid |w| \leq 4 \}$

#f strings in L_2 which are also in L_1.

0	1	2	3	4	
0	1	2	4	8	= 15

a, b, c

$L_1 = \{aa, bb\}$

$L_2 = (a+b)^*$

$L_2 \cap L_1 = L_1$

$L_1 = \{ab, ba\}$

$L_2 = w \in (a+b)^*, \quad |w| \leq 2$

$L_1 \cap L_2 = L_1$

$L_1 = \underbrace{b^* a b^* (a b^* a b^*)^*}_{\#_b\, a \text{ is odd}}$

$0 + 1 + 2 + 4 = 7$

$L_2 = w \in (a+b)^*, \quad |w| \leq 3$

0	1	2	3
ε	a ✓	a a ✗	a a a ✓
	b ✗	a b ✓	a a b ✗
		b a ✓	a b a ✗
		b b ✗	a b b ✓
			b a a ✗
			b a b ✓
			b b a ✓
			b b b ✗
0	1	2	4

(4) $L_1 = a^* b^*$

$\bar{L}_1 = (a+b)^* b a (a+b)^*$

True / False

$\left((a+b)^+\right)^* \cdot (a+b)^+$

(5) $L_1 = (a+b)^+$

A) $L_1^* \cdot L_1 = L_1$ True

B) $L_1^+ \cdot L_1 = L_1$ False

C) $L_1 \cdot L_1 = L_1$ False

D) $\underbrace{\varepsilon \cdot L_1}_{L_1 \cdot \phi = \phi} \cdot \phi = L_1$ ~~True~~ False

Thank You